AIVANCIN

Your next role starts with the work you already do.

Describe what you actually do. AI suggests the matching work activities, you confirm them, and AIVANCIN shows the closest graduate roles and a free first plan.

Team DOVUN · German-Malaysian Institute
SDC Hackathon 2026 · Domain One: The Aiversecity Future-Ready Workforce

The problem

40.7%

of employed Malaysian graduates aged 25 to 34 work in semi-skilled or low-skilled jobs.

751,100 people

Before spending evenings and savings on retraining, they need to know: **which graduate job is closest to the work I already do?**

Source: DOSM, Quarterly Skills-Related Underemployment by Age, Q2 2026 (CC BY 4.0)

Meet Hafiz (a synthetic persona)

27 · Computing diploma · Delivery van driver

His job title says

Vehicle-operator roles

- Light Truck Drivers
- Supervisors of vehicle operators, crane operators

0 of the top 10 roles are in computing

His weekends say

- Fixes friends' laptops and home wifi
- Installs software, cleans out viruses
- Backs up phones and laptops
- Keeps his uncle's shop website updated

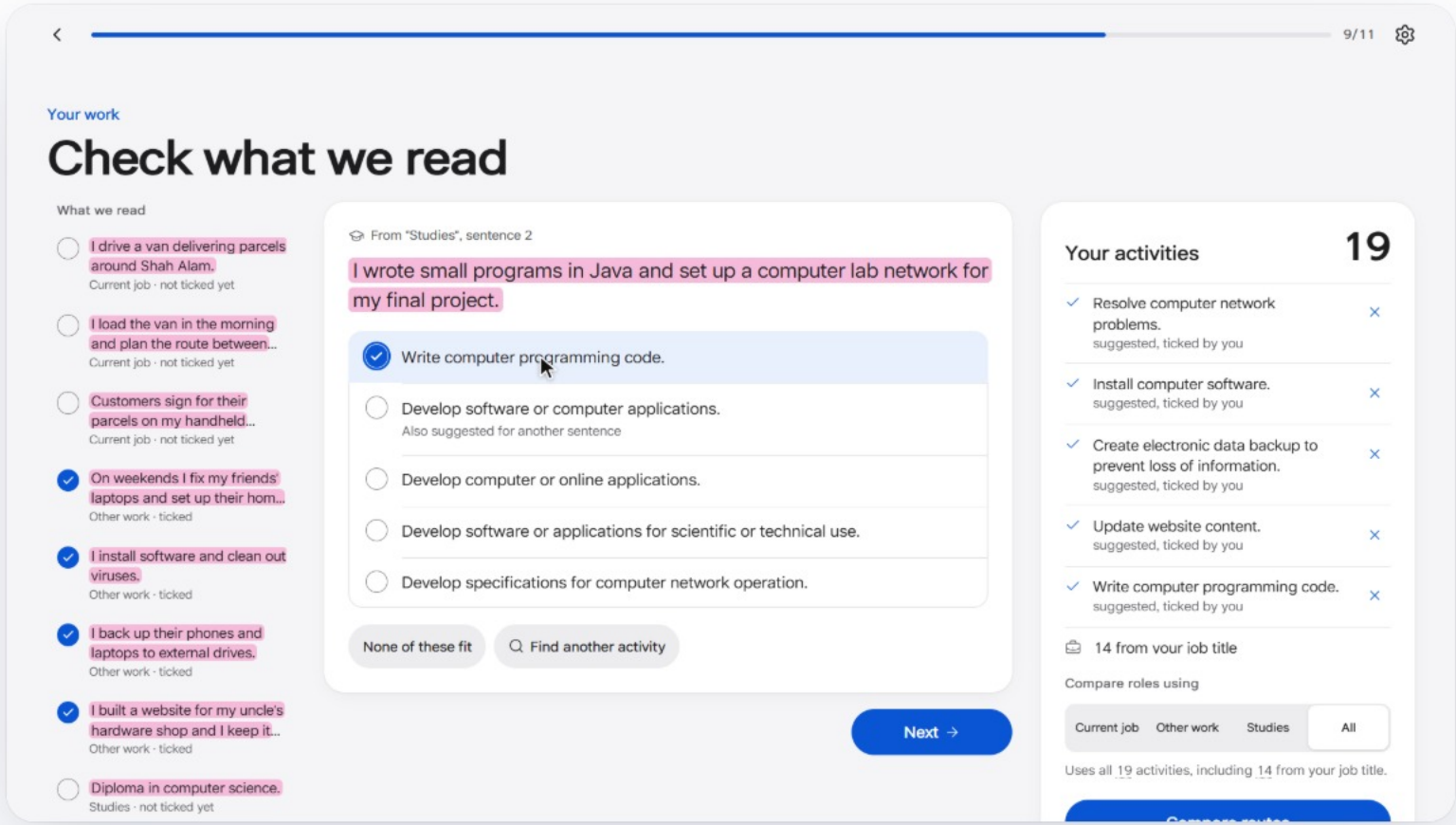
With that work confirmed: at least 6 of the top 10

Job-title tools never see this work. AIVANCIN starts from it.

Live prototype · Describe and confirm

Confirm what is true

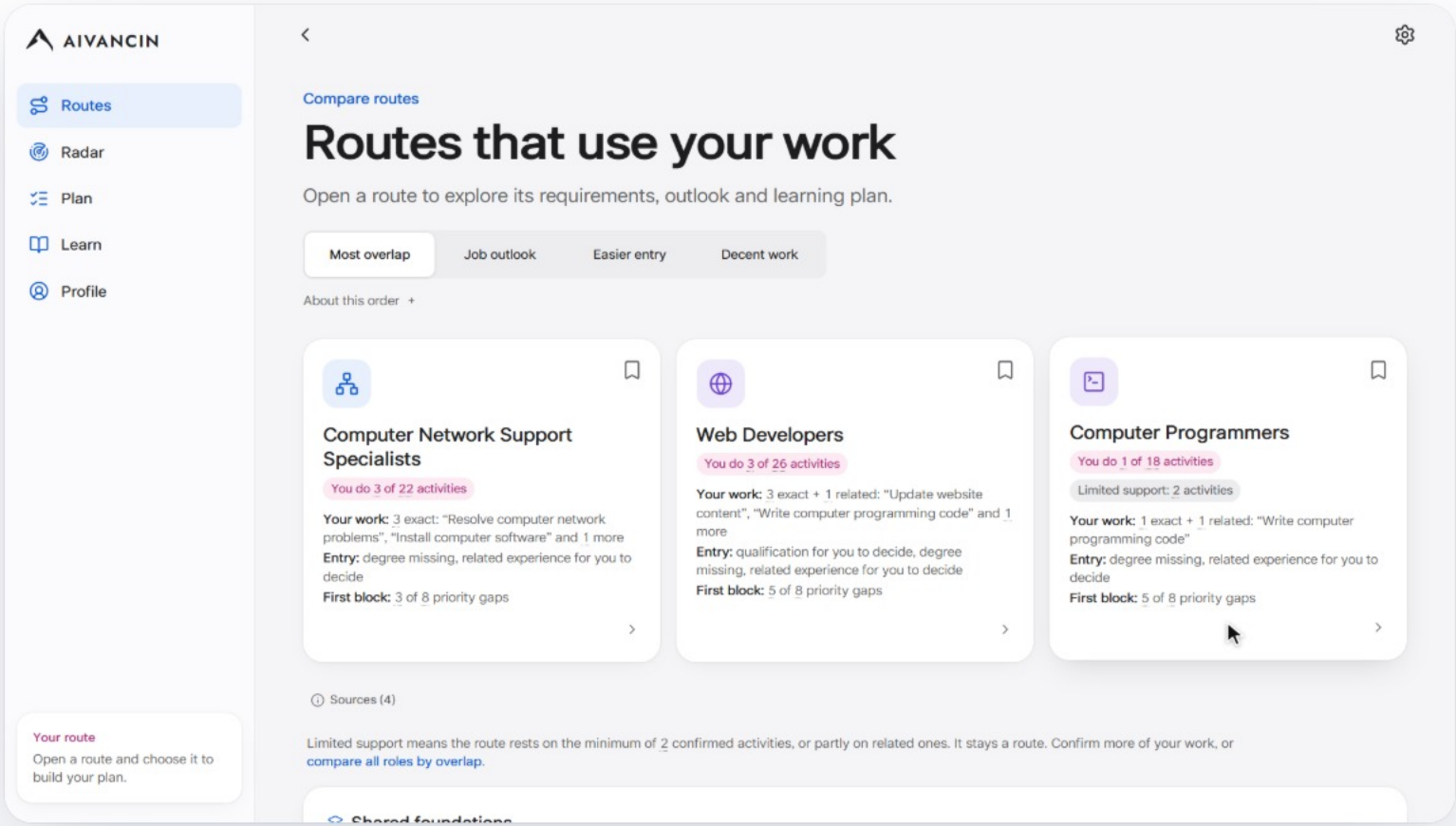
For each sentence, a fine-tuned encoder suggests five of O*NET's 2,087 work activities and highlights the sentence behind them. Nothing counts until the user ticks it.



Live prototype · Compare

Routes that use your work

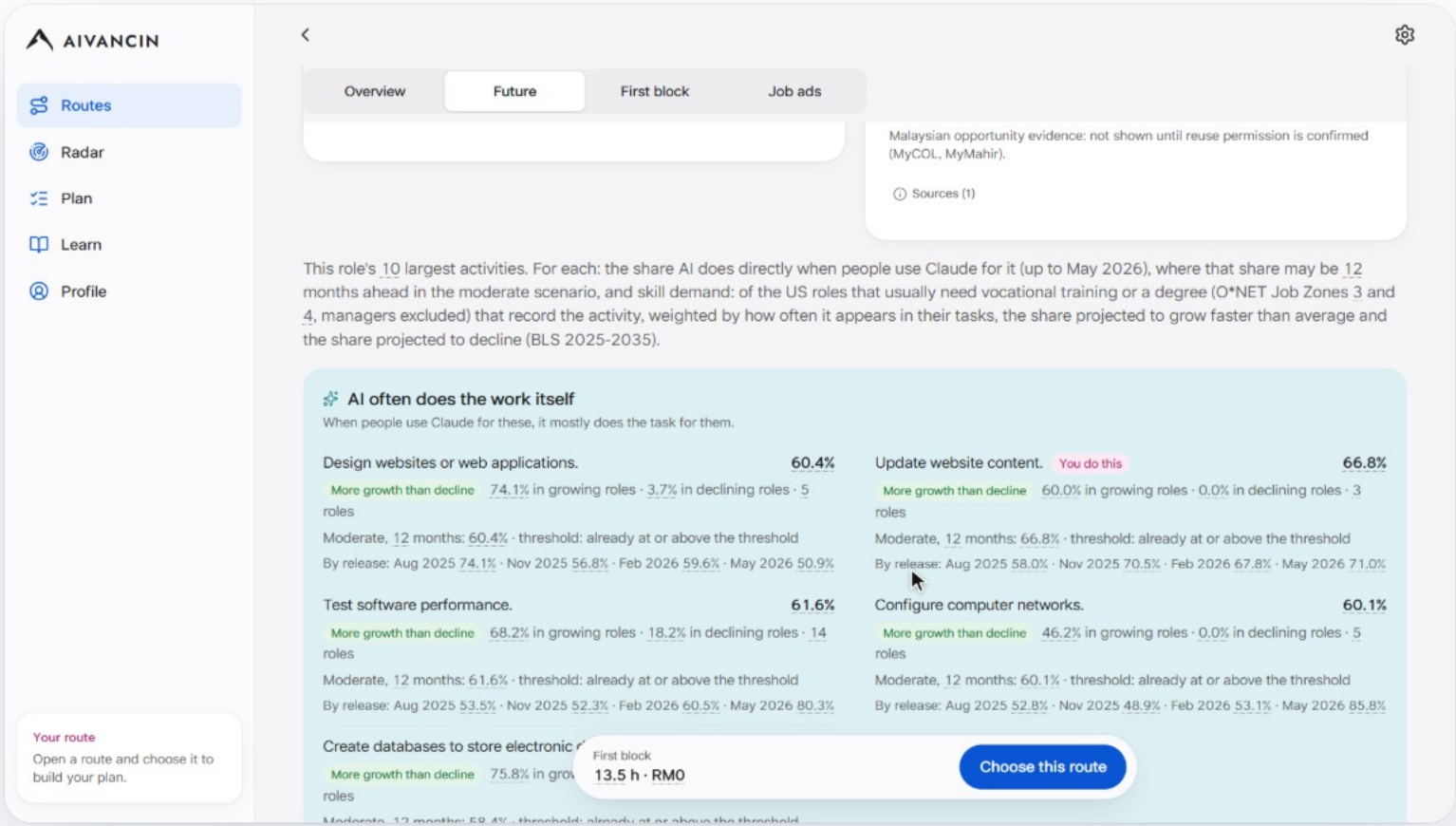
Roles ranked by activity overlap, with the work behind each route, the requirements still open and the gaps the first block covers. Reorder by outlook, entry or decent work.



Live prototype · Route detail

The future of each route

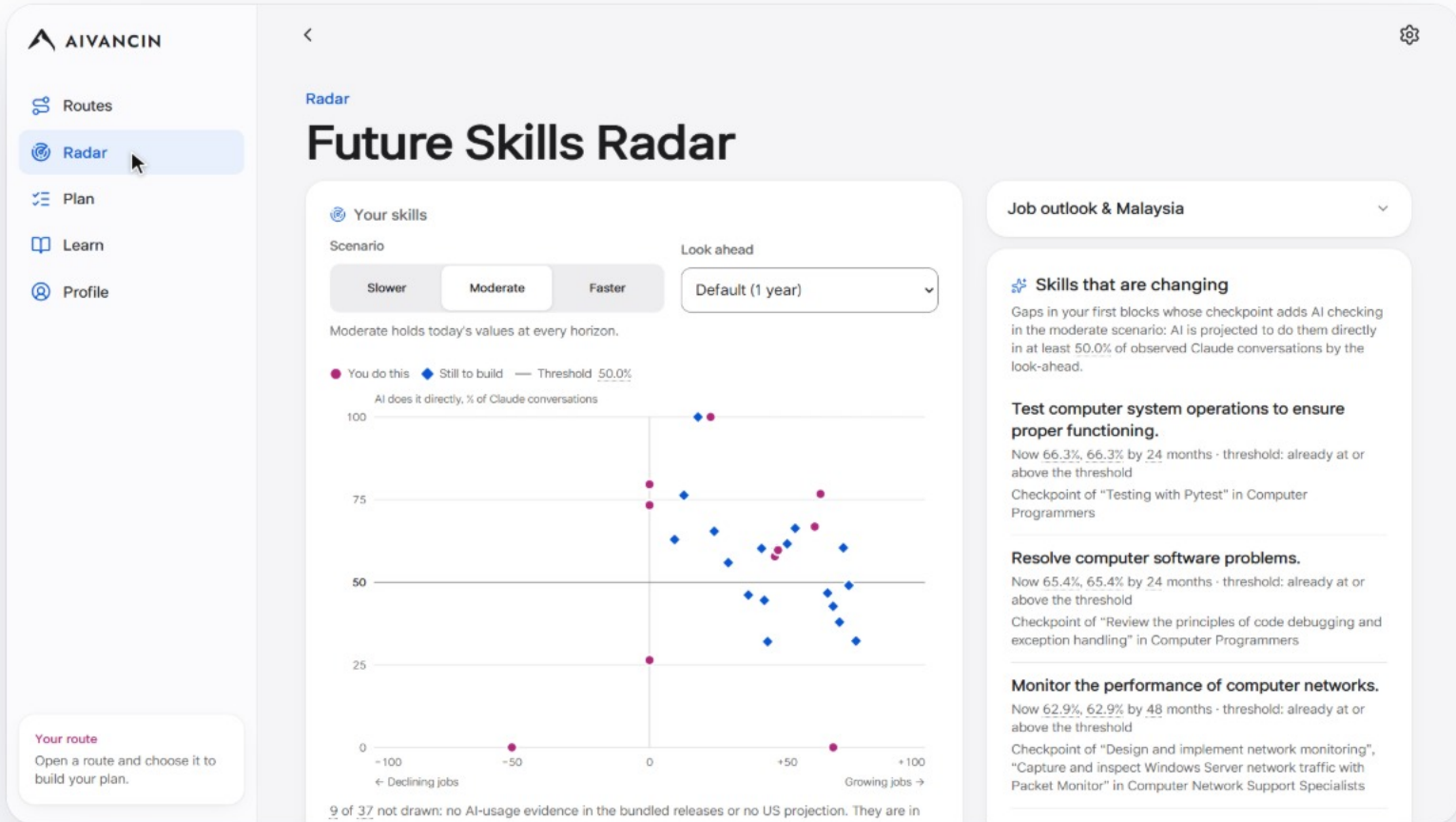
US projections from BLS with their accuracy record, Malaysia's observed DOSM trend, and how much AI already does each activity, under three scenarios.



Live prototype · Radar

Future Skills Radar

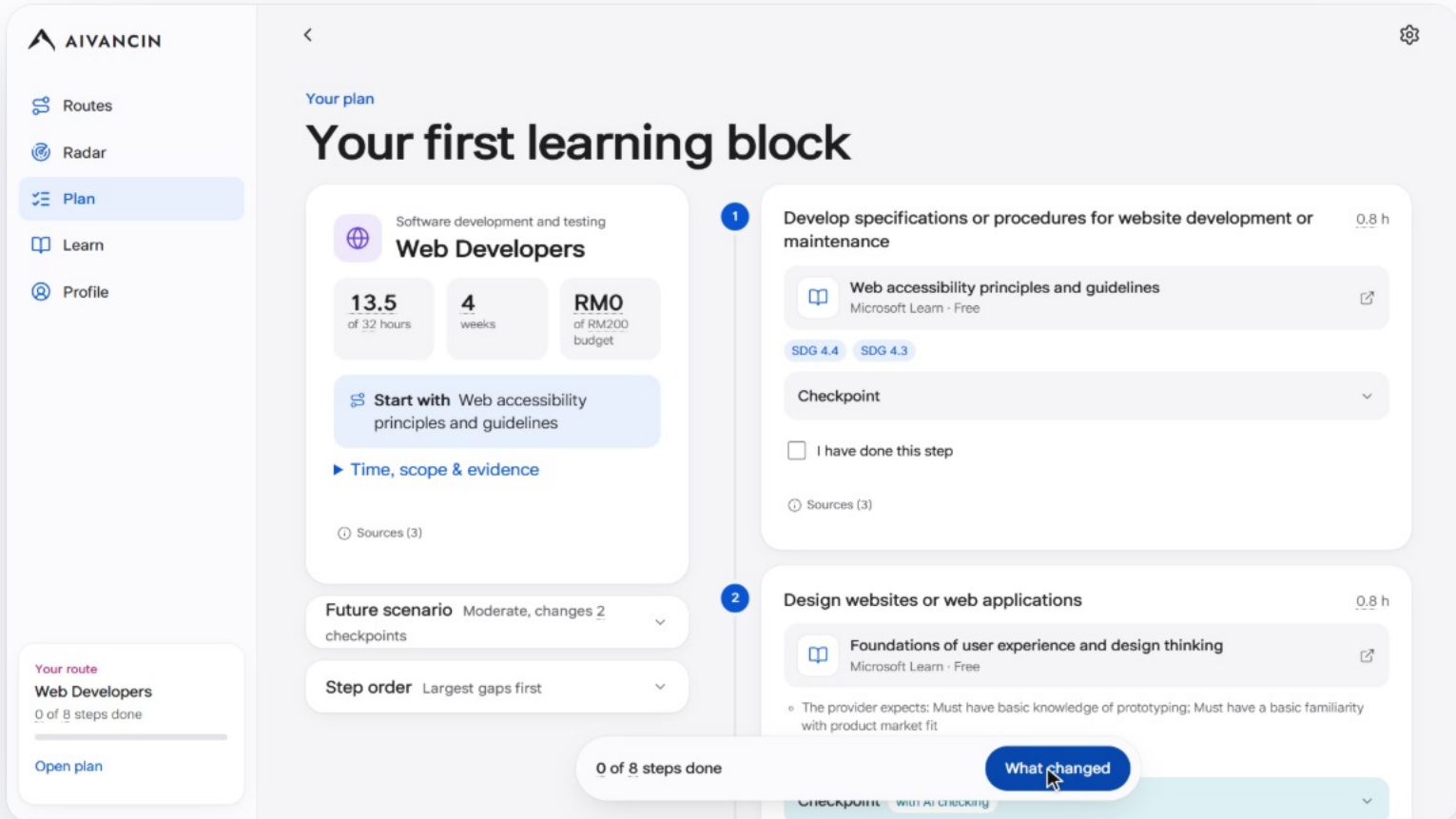
Your skills in pink, your gaps in blue. Across: growing or declining US demand. Up: how much AI does it directly. Scenarios are explicit; estimates carry their backtested error.



Live prototype · Plan

A free first learning block

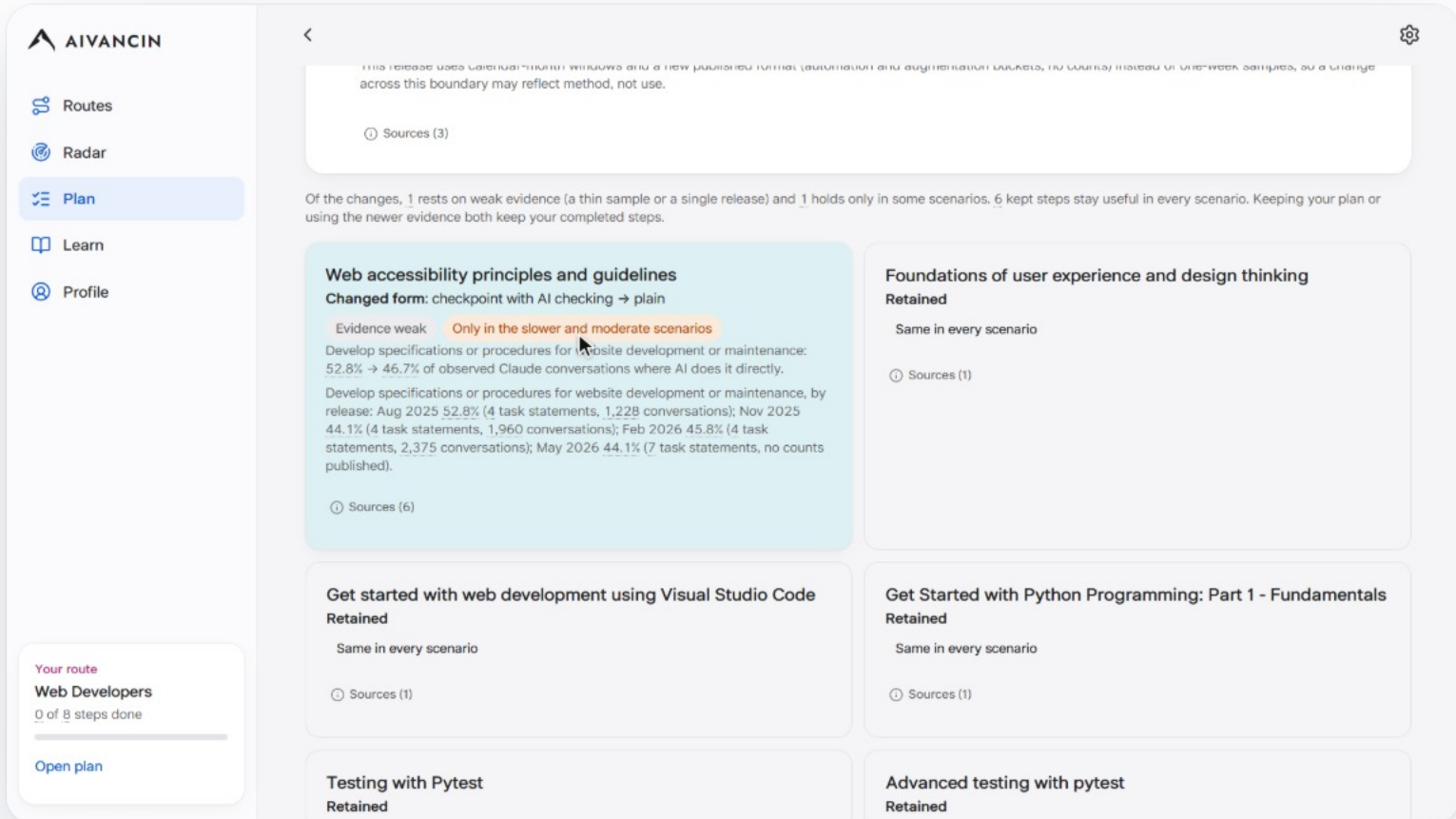
Four weeks within the user's hours and budget: verified free courses in prerequisite order, practice checkpoints and SDG tags with reasons.



Live prototype · What changed

New evidence, your decision

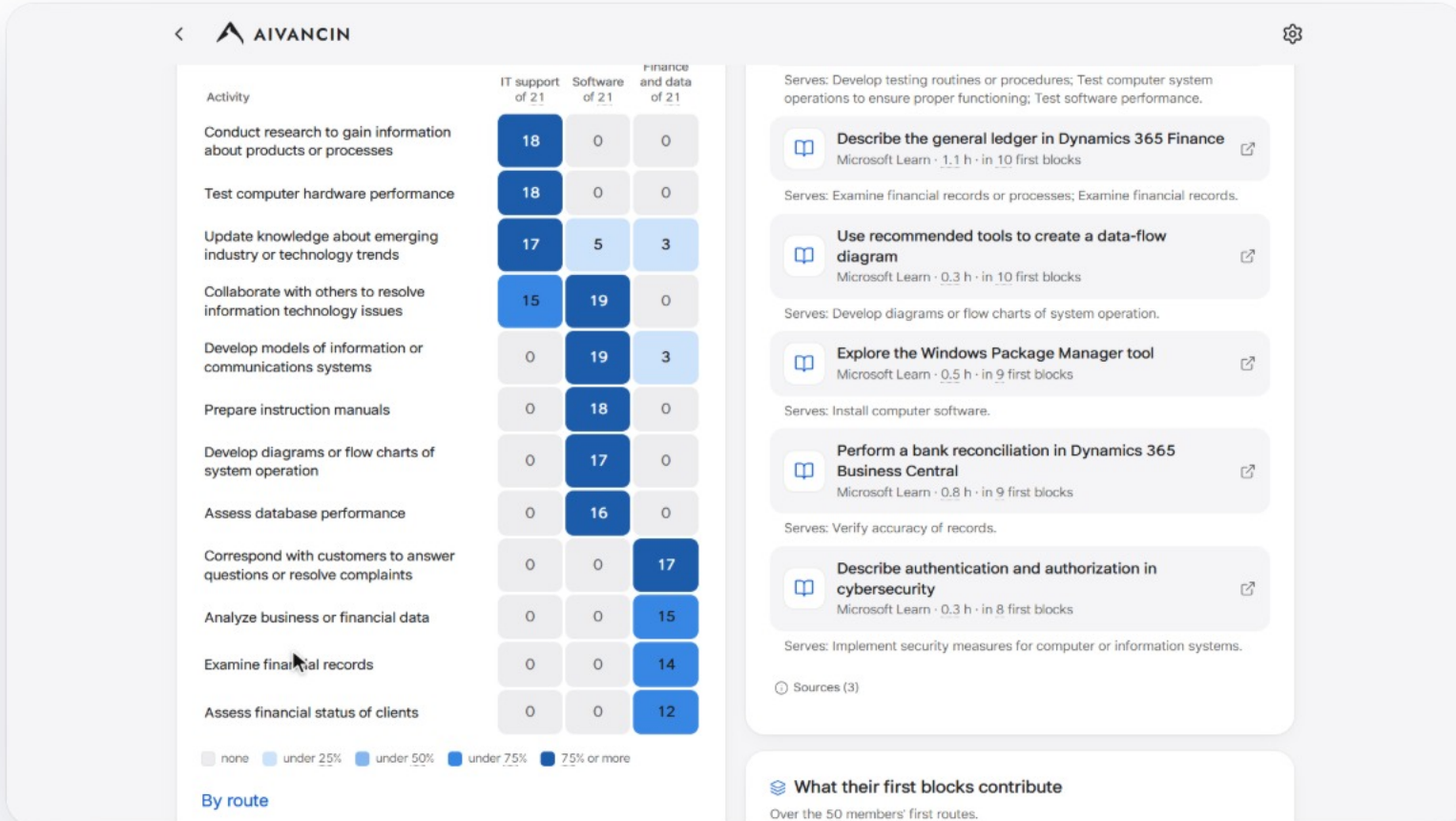
A new AI-usage release is compared with the user's work, target and progress fixed. Each change says whether it rests on weak evidence; the user keeps the plan or accepts the update.



Live prototype · Career centres

A cohort view

Shared priority gaps across a synthetic cohort of 60 graduates, by route family. Counts only, never a score; it changes no one's routes.



Domain One must-haves

All six run in the prototype.

1

Skills profile intake

Four questions in your own words. AI suggests activities; you confirm each one.

2

AI-trend signal layer

Four Anthropic Economic Index releases, BLS outlook and DOSM, each sourced and dated, with a source watch.

3

Future-role matching

Ranked by activity overlap, with requirements, gaps and the work behind each route.

4

Reskilling pathway

An ordered first block of verified free courses, sized to your hours and budget.

5

SDG mapping

Reasoned tags on every step (8.5, 4.4, 4.3, 8.2) and how your path contributes.

6

Handling uncertainty

Slower, moderate and faster scenarios; estimates with their backtested error.

How it works

A small model, checked at every step.

1. **You describe your work**
   plain sentences, your own words
2. **Fine-tuned MiniLM encoder**
   quantized ONNX on the CPU; top five of 2,087 activities
3. **You confirm what is true**
   nothing is accepted silently
4. **Deterministic matching and plans**
   no generated text; same input, same result
5. **Results with sources and dates**
   O*NET, Anthropic Economic Index, BLS, DOSM

RunPod trained

Fine-tuned once on an **RTX 4090** with O*NET task-to-activity pairs. Shipped because it passed a gate fixed before training: **45 of 60** held-out descriptions, against 39. Cost US$0.42.

Replit live

The prototype runs at **aivancin--jadfahmi.replit.app** from bundled, dated, hashed public data. No API keys, no per-request cost.

Evidence

What we measured, including what fails.

2×

relevant roles from the AI chain versus keyword search

32 against 16 of 50 top-five roles, same synthetic text

Pass

45/60

held-out descriptions with a correct activity after fine-tuning

39 before; gate fixed before training

Shipped

35/52

in-scope profiles with two complete routes and blocks

67%, against our 80% target

Not yet met

Hardest risk: matching quality outside IT and finance. **Fallback:** honest exploration states and "no verified resource", never an invented job or course.

Synthetic, predeclared test profiles; no real-user testing yet. 292 automated tests pass. Details in the README.

SDG impact

From the work you do to decent work.

Targets	**SDG 8.5** **SDG 4.4** Decent work for young people; relevant skills for employment. Supporting: 4.3 free learning, 8.2 working with AI.
Baseline	**40.7%** of employed graduates aged 25 to 34 in semi- or low-skilled jobs (751,100; DOSM, Q2 2026)
KPI	Route-and-block coverage: in-scope profiles with two skilled routes, sourced requirements and a free first block. **Now 35 of 52 (67%)**
Phase 2 target	**At least 80%** on a fresh, predeclared profile set; then testing with real underemployed graduates

AIVANCIN

You choose.
AI suggests.

Live prototype aivancin--jadfahmi.replit.app

Code github.com/AIVANCIN/aivancin

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